有限 SYNTOMIC 态射的微分形式迹

Counting scope. The primary problem is Tag 0FLB, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题构造与基变换相容、在拉回微分形式上限制为普通迹的微分形式迹映射。证明先用 upstairs 映射及 Tate 同构获得 Noetherian 情形，再在通用有限 syntomic 模型上证明唯一性并以 Noetherian 逼近扩展至任意基底。构造所用核心映射是先修输入，但唯一性、极限定义及基变换相容仍由本证明组织，判 H2。.

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Lemma 0.1. *There exists a unique rule that to every locally quasi-finite syntomic morphism of schemes $f : Y \to X$ assigns $\mathcal{O}_Y$-module maps*

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

for $p \geq 0$ satisfying (item degree zero (Tag 0H9C)), (item multiplicative (Tag 0H9D)), and (item base change (Tag 0H9E)). In particular, the composition of $c^p_{Y/X}$ with $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ is multiplication by $\delta(NL_{Y/X})$.

证明. This proof is very similar to the proof of Discriminants, Proposition 0.2 and we suggest the reader look at that proof first.

Let us reformulate the statement. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $f : Y \to X$ of schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow[b]{} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The lemma means that for every object Y/X of $\mathcal{C}$ we have maps $c^p_{Y/X}$, $p \geq 0$ with properties (item degree zero (Tag 0H9C)) and (item multiplicative (Tag 0H9D)), and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have that $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma lemma base change Garel upstairs (Tag 0H9G).

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine open $V \subset Y$ and $U \subset X$ with $f(V) \subset U$ such that there exists some maps

$$\Omega^p_{Y/\mathbf{Z}}|_V \longrightarrow \left(f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})\right)|_V$$

with properties (item degree zero (Tag 0H9C)) and (item multiplicative (Tag 0H9D)). This follows from picking affine opens as in Discriminants, Lemma lemma syntomic quasi finite (Tag 0BWE) part (5) and then using the construction in Remark remark local description (Tag 0FL9).

Note that the étale locus of f is exactly the set of points where $\delta(NL_{Y/X})$ does not vanish, see discussion in Discriminants, Section section tate map (Tag 0FKB). In particular $f^*\Omega^p_{X/\mathbf{Z}} \to \Omega^p_{Y/\mathbf{Z}}$ becomes an isomorphism after inverting $\delta(NL_{Y/X})$. We conclude that if $\Omega^p_{X/\mathbf{Z}}$ is finite locally free and the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero, then the local maps constructed in the previous paragraph are unique and automatically glue!

Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that

(1) $\Omega_{X/\mathbf{Z}}$ is locally free, and
(2) the annihilator of $\delta(NL_{Y/X})$ in $\mathcal{O}_Y$ is zero.

By the remarks in the previous paragraph, we see that for any object Y/X of $\mathcal{C}_{nice}$ we have unique maps $c^p_{Y/X}$, $p \geq 0$ satisfying conditions (item degree zero (Tag 0H9C)) and (item multiplicative (Tag 0H9D)). If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}_{nice}$, then the maps $c^p_{Y/X}$ and $c^p_{Y'/X'}$ are compatible as in Lemma lemma base change Garel upstairs (Tag 0H9G): namely, locally there do exist compatible maps by Lemma lemma existence base change Garel upstairs (Tag 0H9H) and the uniqueness just mentioned shows these maps agree with $c^p_{Y/X}$ and $c^p_{Y'/X'}$. In other words, we have solved the problem on the full subcategory $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c^p_{Y/X}$ the solution we've just found.

In fact, more generally, suppose we have a morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ such that Y/X is an object of $\mathcal{C}_{nice}$. The same argument, but this time using the uniqueness in Lemma lemma base change Garel upstairs (Tag 0H9G), tells us there are maps $c^p_{Y'/X'}$, $p \geq 0$ satisfying conditions (item degree zero (Tag 0H9C)) and (item multiplicative (Tag 0H9D)) compatible with the already constructed maps $c^p_{Y/X}$. Let us call this the base change $(b/a)^*c^p_{Y/X}$.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** The two base changes $(b_i/a_i)^*c^p_{Y_i/X_i}$, $i = 1, 2$ are equal. We will first show that the claim implies the lemma and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Discriminants, Example example universal quasi finite syntomic (Tag 0FK8). Then $Y_{n,d}$ and $Y_{n,d}$ are irreducible schemes of finite type and smooth over $\mathbf{Z}$. Namely, $X_{n,d}$ is a spectrum of a polynomial ring over $\mathbf{Z}$ and $Y_{n,d}$ is an open subscheme of such. The morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Discriminants, Lemma lemma universal quasi finite syntomic etale (Tag 0FK9). Thus $\delta(NL_{Y_{n,d}/X_{n,d}})$ is nonzero: for example we have the local description of $\delta(NL_{Y/X})$ in Discriminants, Remark remark local description delta (Tag 0FKC) and we have the local description of étale morphisms in Morphisms, Lemma lemma etale at point (Tag 02GU) part (8). Now a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Discriminants, Lemma lemma locally comes from universal (Tag 0FKA) we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Then we have the base change $c^p_{V/U} = (b/a)^*c^p_{Y_{n,d}/X_{n,d}}$. The claim guarantees these locally constructed maps $c^p_{V/U}$ glue! Thus we get a well defined global

maps $c^p_{Y/X}$ with properties (item degree zero (Tag 0H9C)) and (item multiplicative (Tag 0H9D)). Finally, let $b/a : Y'/X' \to Y/X$ be an arbitray morphism of $\mathcal{C}$. We have the maps $c^p_{Y'/X'}$, $p \geq 0$ and $c^p_{Y/X}$, $p \geq 0$ constructed in this paragraph. To check they are compatible as in Lemma lemma base change Garel upstairs (Tag 0H9G) we may work locally on Y'/X' and Y/X. Thus we may assume that there exists a morphism $Y/X \to Y_{n,d}/X_{n,d}$. By construction, both the family $c^p_{Y'/X'}$ and the family $c^p_{X/Y}$ are compatible with the map towards $Y_{n,d}/X_{n,d}$. It follows easily from this (and the uniqueness in Lemma lemma base change Garel upstairs (Tag 0H9G)) that $c^p_{Y'/X'}$ is compatible with $c^p_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma lemma standard open two affines (Tag 0H9B). Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A'_1 = A_1[x_\alpha] \to A \quad \text{and} \quad A'_2 = A_2[y_\beta] \to A$$

of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A'_i \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

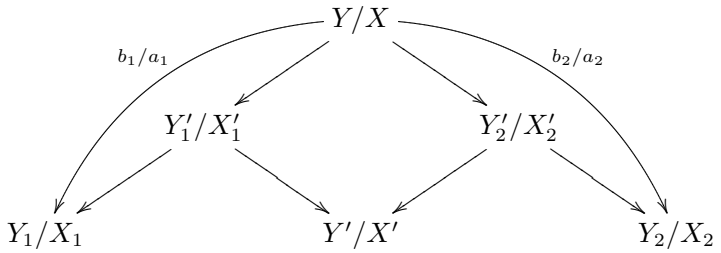
$$\begin{array}{ccccc} (A'_1 \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A'_2 \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A'_1 & \longrightarrow & A & \longleftarrow & A'_2 \end{array}$$

By construction the two squares are cocartesian. Next, we consider the ring map

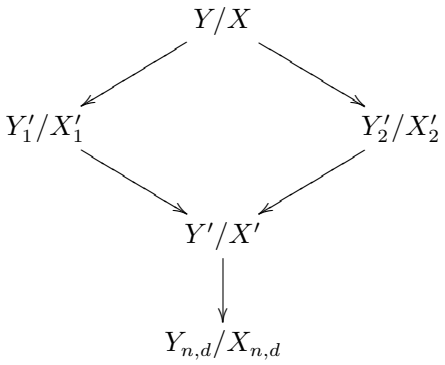
$$A' = A'_1 \times_A A'_2 \longrightarrow B' = (A'_1 \otimes_{A_1} B_1)_{h_1} \times_B (A'_2 \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma lemma module over fibre product (Tag 07RU) we have $A'_1 \otimes_{A'} B' = (A'_1 \otimes_{A_1} B_1)_{h_1}$ and $A'_2 \otimes_{A'} B' = (A'_2 \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Discriminants, Lemma lemma syntomic quasi finite (Tag 0BWE)). By More on Algebra, Lemma lemma properties algebras over fibre product (Tag 08KQ) the ring map $A' \to B'$ is flat and of finite presentation. Thus by Discriminants, Lemma lemma

syntomic quasi finite (Tag 0BWE) part (3) we conclude that Y'/X' is an object of $\mathcal{C}$. Consider now the commutative diagram



where Y_i'/X_i' corresponds to $A_i' \to (A_i' \otimes_{A_i} B_i)_{h_i}$. Note that Y_i'/X_i' is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an infinite dimensional affine space with Y_i/X_i; small detail omitted. In particular, the pullback of $c^p_{Y_i/X_i}$ via (b_i/a_i) is the same as the pullback of $c^p_{Y_i'/X_i'}$ via $Y/X \to Y_i'/X_i'$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$, but this is almost never the case. Namely, then pulling back $c^p_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X_1', Y_1', X_2', Y_2', X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c^p_{Y_{n,d}/X_{n,d}}$. This finishes the proof. □

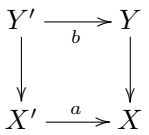
Proposition 0.2. *There exists a unique rule that to every locally quasi-finite syntomic morphism of locally Noetherian schemes $Y \to X$ assigns an isomorphism*

$$c_{Y/X} : \det(NL_{Y/X}) \longrightarrow \omega_{Y/X}$$

satisfying the following two properties

(1) *the section $\delta(NL_{Y/X})$ is mapped to $\tau_{Y/X}$, and*
(2) *the rule is compatible with restriction to opens and with base change.*

证明. Let us reformulate the statement of the proposition. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $Y \to X$ of locally Noetherian schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams



which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The proposition means that for every object Y/X of $\mathcal{C}$ we have an isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$ and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have $b^* c_{Y/X} = c_{Y'/X'}$ via the identifications $b^* \det(NL_{Y/X}) = \det(NL_{Y'/X'})$ and $b^*\omega_{Y/X} = \omega_{Y'/X'}$ described above.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine opens $V \subset Y$ and $U \subset X$ with $y \in V$ and $f(V) \subset U$ such that there exists some isomorphism

$$\det(NL_{Y/X})|_V \longrightarrow \omega_{Y/X}|_V$$

mapping $\delta(NL_{Y/X})|_V$ to $\tau_{Y/X}|_V$. This follows from picking affine opens as in Lemma lemma syntomic quasi finite (Tag 0BWE) part (5), the affine local description of $\delta(NL_{Y/X})$ in Remark remark local description delta (Tag 0FKC), and Lemma lemma different quasi finite complete intersection (Tag 0BWH). If the annihilator of the section $\tau_{Y/X}$ is zero, then these local maps are unique and automatically glue. Hence if the annihilator of $\tau_{Y/X}$ is zero, then there is a unique isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$. If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}$ and the annihilator of $\tau_{Y'/X'}$ is zero as well, then $b^*c_{Y/X}$ is the unique isomorphism $c_{Y'/X'} : \det(NL_{Y'/X'}) \to \omega_{Y'/X'}$ with $c_{Y'/X'}(\delta(NL_{Y'/X'})) = \tau_{Y'/X'}$. This follows formally from the fact that $b^*\delta(NL_{Y/X}) = \delta(NL_{Y'/X'})$ and $b^*\tau_{Y/X} = \tau_{Y'/X'}$.

We can summarize the results of the previous paragraph as follows. Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that the annihilator of $\tau_{Y/X}$ is zero. Then we have solved the problem on $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c_{Y/X}$ the solution we've just found.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** $b_1^* c_{Y_1/X_1} = b_2^* c_{Y_2/X_2}$. We will first show that the claim implies the proposition and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Example example universal quasi finite syntomic (Tag 0FK8). Then $Y_{n,d}$ is an irreducible regular scheme and the morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Lemma lemma universal quasi finite syntomic etale (Tag 0FK9). Thus $\tau_{Y_{n,d}/X_{n,d}}$ is nonzero for example by Lemma lemma different ramification (Tag 0BW9). Now a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Lemma lemma locally comes from universal (Tag 0FKA) we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Thus we can pullback the canonical morphism $c_{Y_{n,d}/X_{n,d}}$ constructed above by b to V. The claim guarantees these local isomorphisms glue! Thus we get a well defined global isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$. If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}$, then the claim also implies that the similarly constructed map $c_{Y'/X'}$ is the pullback by b of the locally constructed map $c_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma lemma standard open two affines (Tag 0H9B). Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A_1' = A_1[x_\alpha] \to A \quad \text{and} \quad A_2' = A_2[y_\beta] \to A$$

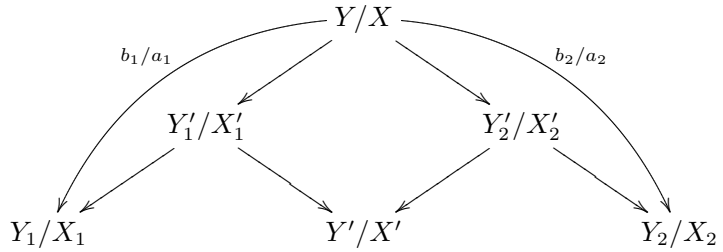
of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A_i' \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

$$\begin{array}{ccccc} (A_1' \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A_2' \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A_1' & \longrightarrow & A & \longleftarrow & A_2' \end{array}$$

By construction the two squares are cocartesian. Next, we consider the ring map

$$A' = A_1' \times_A A_2' \longrightarrow B' = (A_1' \otimes_{A_1} B_1)_{h_1} \times_B (A_2' \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma lemma module over fibre product (Tag 07RU) we have $A_1' \otimes_{A'} B' = (A_1' \otimes_{A_1} B_1)_{h_1}$ and $A_2' \otimes_{A'} B' = (A_2' \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Lemma lemma syntomic quasi finite (Tag 0BWE)). By More on Algebra, Lemma lemma properties algebras over fibre product (Tag 08KQ) the ring map $A' \to B'$ is flat and of finite presentation. Thus by Lemma lemma syntomic quasi finite (Tag 0BWE) part (3) we conclude that $Y' \to X'$ is locally quasi-finite and syntomic. Thus we obtain the commutative diagram

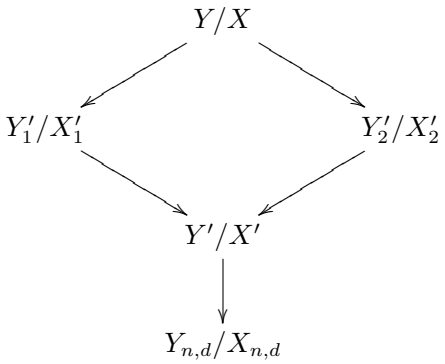


where Y_i'/X_i' corresponds to $A_i' \to (A_i' \otimes_{A_i} B_i)_{h_i}$.

In this paragraph, we fix the issue that A' and A_i' may not be Noetherian by a limit argument; we suggest the reader skip this. We can find a finitely generated $\mathbf{Z}$-subalgebra $A'' \subset A'$ and a quasi-finite and syntomic ring map $A'' \to B''$ such that $B' = A' \otimes_{A''} B''$. This follows from Limits, Lemmas lemma descend finite presentation (Tag 01ZM), lemma descend syntomic (Tag 0C3L), and lemma descend quasi finite (Tag 094M). Then the ring maps $A'' \to A_1' = A_1[x_\alpha]$ will factor through a finite polynomial

subalgebra $A_i[x_1, \ldots, x_n$ and the ring map $A'' \to A'_2 = A_2[y_\beta]$ will factor through $A_2[y_1, \ldots, y_m]$. After increasing the sets of variables, we may also assume that $h_1 \in A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1$ and $h_2 \in A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2$. Increasing the sets of variables one more time, we can assume that the maps $B'' \to (A'_i \otimes_{A_i} B_i)_{h_i}$ factor through $(A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1)_{h_1}$ and $(A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2)_{h_2}$. Replacing Y', X', Y'_1, X'_1, Y'_2, X'_2 by the spectra of B'', A'', $(A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1)_{h_1}$, $A_1[x_1, \ldots, x_n]$, $(A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2)_{h_2}$, $A_2[y_1, \ldots, y_m]$, we see that we get a diagram as above where all schemes are Noetherian.

Note that Y'_i/X'_i is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an affine space with Y_i/X_i. In particular, the pullback of c_{Y_i/X_i} via (b_i/a_i) is the same as the pullback of $c_{Y'_i/X'_i}$ via $Y/X \to Y'_i/X'_i$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$ (but this is likely not the case). Namely, then pulling back $c_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X'_1, Y'_1, X'_2, Y'_2, X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c_{Y_{n,d}/X_{n,d}}$. This finishes the proof. □

1. Finite morphisms

In this section we collect some observations about the constructions in the previous sections for finite morphisms. Let $f : Y \to X$ be a finite morphism of locally Noetherian schemes. Let $\omega_{Y/X}$ be as in Remark remark relative dualizing for quasi finite (Tag 0BVG).

The first remark is that

$$f_*\omega_{Y/X} = \mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X)$$

as sheaves of $f_*\mathcal{O}_Y$-modules. Since f is affine, this formula uniquely characterizes $\omega_{Y/X}$, see Morphisms, Lemma lemma affine equivalence modules (Tag 01SB). The formula holds because for $\mathrm{Spec}(A) = U \subset X$ affine open, the inverse image $V = f^{-1}(U)$ is the spectrum of a finite A-algebra B and hence

$$H^0(U, f_*\omega_{Y/X}) = H^0(V, \omega_{Y/X}) = \omega_{B/A} = \mathrm{Hom}_A(B, A) = H^0(U, \mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X))$$

by construction. In particular, we obtain a canonical evaluation map

$$f_*\omega_{Y/X} \longrightarrow \mathcal{O}_X$$

which is given by evaluation at 1 if we think of $f_*\omega_{Y/X}$ as the sheaf $\mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X)$.

The second remark is that using the evaluation map we obtain canonical identifications

$$\mathrm{Hom}_Y(\mathcal{F}, f^*\mathcal{G} \otimes_{\mathcal{O}_Y} \omega_{Y/X}) = \mathrm{Hom}_X(f_*\mathcal{F}, \mathcal{G})$$

functorially in the quasi-coherent module $\mathcal{F}$ on Y and the finite locally free module $\mathcal{G}$ on X. If $\mathcal{G} = \mathcal{O}_X$ this follows immediately from the above and Algebra, Lemma lemma adjoint hom restrict (Tag 08YP). For general $\mathcal{G}$ we can use the same lemma and the isomorphisms

$$f_*(f^*\mathcal{G} \otimes_{\mathcal{O}_Y} \omega_{Y/X}) = \mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{O}_X) = \mathcal{H}om_{\mathcal{O}_X}(f_*\mathcal{O}_Y, \mathcal{G})$$

of $f_*\mathcal{O}_Y$-modules where the first equality is the projection formula (Cohomology, Lemma lemma projection formula (Tag 01E8)). An alternative is to prove the formula affine locally by direct computation.

The third remark is that if f is in addition flat, then the composition

$$f_*\mathcal{O}_Y \xrightarrow{f_*\tau_{Y/X}} f_*\omega_{Y/X} \longrightarrow \mathcal{O}_X$$

is equal to the trace map Trace_f discussed in Section section discriminant (Tag 0BVH). This follows immediately by looking over affine opens.

The fourth remark is that if f is flat and X Noetherian, then we obtain

$$\mathrm{Hom}_Y(K, Lf^*M \otimes_{\mathcal{O}_Y} \omega_{Y/X}) = \mathrm{Hom}_X(Rf_*K, M)$$

for any K in $D_{QCoh}(\mathcal{O}_Y)$ and M in $D_{QCoh}(\mathcal{O}_X)$. This follows from the material in Duality for Schemes, Section section proper flat (Tag 0E4H), but can be proven directly in this case as follows. First, if X is affine, then it holds by Dualizing Complexes, Lemmas lemma right adjoint (Tag 0A70) and lemma RHom is tensor special (Tag 0BZE)[1] and Derived Categories of Schemes, Lemma lemma affine compare bounded (Tag 06Z0). Then we can use the induction principle (Cohomology of Schemes, Lemma lemma induction principle (Tag 08DR)) and Mayer-Vietoris (in the form of Cohomology, Lemma lemma mayer vietoris hom (Tag 08BW)) to finish the proof.

Primary problem: Tag 0FLB

Lemma 1.1. *There exists a unique rule that to every finite syntomic morphism of schemes $f : Y \to X$ assigns $\mathcal{O}_X$-module maps*

$$\Theta^p_{Y/X} : f_*\Omega^p_{Y/\mathbf{Z}} \longrightarrow \Omega^p_{X/\mathbf{Z}}$$

satisfying the following properties

1. *the composition with $\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \to f_*\Omega^p_{Y/\mathbf{Z}}$ is equal to $id \otimes \mathrm{Trace}_f$ where $\mathrm{Trace}_f : f_*\mathcal{O}_Y \to \mathcal{O}_X$ is the map from Discriminants, Section section discriminant (Tag 0BVH),*
2. *the rule is compatible with base change.*

证明. First, assume that X is locally Noetherian. By Lemma 0.1 we have a canonical map

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

By Discriminants, Proposition 0.2 we have a canonical isomorphism

$$c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$$

[1] There is a simpler proof of this lemma in our case.

mapping $\delta(NL_{Y/X})$ to $\tau_{Y/X}$. Combined these maps give

$$c^p_{Y/X} \otimes c_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \omega_{Y/X}$$

By Discriminants, Section 1 this is the same thing as a map

$$\Theta^p_{Y/X} : f_*\Omega^p_{Y/\mathbf{Z}} \longrightarrow \Omega^p_{X/\mathbf{Z}}$$

Recall that the relationship between $c^p_{Y/X} \otimes c_{Y/X}$ and $\Theta^p_{Y/X}$ uses the evaluation map $f_*\omega_{Y/X} \to \mathcal{O}_X$ which sends $\tau_{Y/X}$ to $\mathrm{Trace}_f(1)$, see Discriminants, Section 1. Hence property (1) holds. Property (2) holds for base changes by $X' \to X$ with X' locally Noetherian because both $c^p_{Y/X}$ and $c_{Y/X}$ are compatible with such base changes. For $f : Y \to X$ finite syntomic and X locally Noetherian, we will continue to denote $\Theta^p_{Y/X}$ the solution we've just found.

Uniqueness. Suppose that we have a finite syntomic morphism $f : Y \to X$ such that X is smooth over $\mathrm{Spec}(\mathbf{Z})$ and f is étale over a dense open of X. We claim that in this case $\Theta^p_{Y/X}$ is uniquely determined by property (1). Namely, consider the maps

$$\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \to f_*\Omega^p_{Y/\mathbf{Z}} \to \Omega^p_{X/\mathbf{Z}}$$

The sheaf $\Omega^p_{X/\mathbf{Z}}$ is torsion free (by the assumed smoothness), hence it suffices to check that the restriction of $\Theta^p_{Y/X}$ is uniquely determined over the dense open over which f is étale, i.e., we may assume f is étale. However, if f is étale, then $f^*\Omega_{X/\mathbf{Z}} = \Omega_{Y/\mathbf{Z}}$ hence the first arrow in the displayed equation is an isomorphism. Since we've pinned down the composition, this guarantees uniqueness.

Let $f : Y \to X$ be a finite syntomic morphism of locally Noetherian schemes. Let $x \in X$. By Discriminants, Lemma lemma locally comes from universal finite (Tag 0FL4) we can find $d \geq 1$ and a commutative diagram

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & V_d \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & U_d \end{array}$$

such that $x \in U \subset X$ is open, $V = f^{-1}(U)$ and $V = U \times_{U_d} V_d$. Thus $\Theta^p_{Y/X}|_V$ is the pullback of the map $\Theta^p_{V_d/U_d}$. However, by the discussion on uniqueness above and Discriminants, Lemmas lemma universal finite syntomic smooth (Tag 0FL1) and lemma universal finite syntomic etale (Tag 0FL2) the map $\Theta^p_{V_d/U_d}$ is uniquely determined by the requirement (1). Hence uniqueness holds.

At this point we know that we have existence and uniqueness for all finite syntomic morphisms $Y \to X$ with X locally Noetherian. We could now give an argument similar to the proof of Lemma 0.1 to extend to general X. However, instead it is possible to directly use absolute Noetherian approximation to finish the proof. Namely, to construct $\Theta^p_{Y/X}$ it suffices to do so Zariski locally on X (provided we also show the uniqueness). Hence we may assume X is affine (small detail omitted). Then we can write $X = \lim_{i \in I} X_i$ as the limit over a directed set I of Noetherian affine schemes. By Algebra, Lemma lemma colimit category fp algebras (Tag 05N9) we can find $0 \in I$ and a finitely presented morphism of affines $f_0 : Y_0 \to X_0$ whose base change to X is $Y \to X$. After increasing 0 we may assume $Y_0 \to X_0$ is finite and syntomic, see Algebra, Lemma lemma colimit lci (Tag 0C33) and lemma colimit finite (Tag 07RG). For $i \geq 0$ also the base change $f_i : Y_i = Y_0 \times_{X_0} X_i \to X_i$ is finite syntomic. Then

$$\Gamma(X, f_*\Omega^p_{Y/\mathbf{Z}}) = \Gamma(Y, \Omega^p_{Y/\mathbf{Z}}) = \mathrm{colim}_{i \geq 0} \Gamma(Y_i, \Omega^p_{Y_i/\mathbf{Z}}) = \mathrm{colim}_{i \geq 0} \Gamma(X_i, f_{i,*}\Omega^p_{Y_i/\mathbf{Z}})$$

Hence we can (and are forced to) define $\Theta^p_{Y/X}$ as the colimit of the maps $\Theta^p_{Y_i/X_i}$. This map is compatible with any cartesian diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

with X' affine as we know this for the case of Noetherian affine schemes by the arguments given above (small detail omitted; hint: if we also write $X' = \lim_{j \in J} X'_j$ then for every $i \in I$ there is a $j \in J$ and a morphism $X'_j \to X_i$ compatible with the morphism $X' \to X$). This finishes the proof. $\square$